

Machine Learning Classification Project Guide

Files Included:

1. task1.py
2. (Optional) Jupyter Notebook version

Project Description

This project builds and evaluates a Machine Learning Classification Model using the Breast Cancer Dataset from scikit-learn.

Algorithms Used:

1. Logistic Regression
2. Random Forest Classifier

Metrics Evaluated:

Accuracy, Precision, Recall, F1 Score, ROC-AUC

How to Run the Project

Step 1: Install Python 3.12 or later and verify using: `python --version`

Step 2: Install libraries:

```
python -m pip install scikit-learn pandas matplotlib jupyter ipykernel
```

Step 3: Run the project:

```
python task1.py
```

Expected Output: Cross Validation Scores, Accuracy, Precision, Recall, F1 Score, ROC-AUC Score, and a Confusion Matrix.

Create Jupyter Notebook (Optional)

Install Python and Jupyter extensions in VS Code, create a notebook, paste code into cells, select Python 3.12 kernel, and click Run All.

Project Requirements Satisfied

- ✓ Data Preprocessing
- ✓ Train/Test Split
- ✓ Cross Validation
- ✓ Logistic Regression
- ✓ Random Forest
- ✓ Accuracy
- ✓ Precision
- ✓ Recall
- ✓ F1 Score
- ✓ ROC-AUC
- ✓ Confusion Matrix Plot

Results

Logistic Regression CV: ~95%

Random Forest CV: ~95%

Accuracy: 96.49%

Precision: 95.89%

Recall: 98.59%

F1 Score: 97.22%

ROC-AUC: 99.71%

Conclusion: Random Forest achieved excellent classification performance and was selected as the final model.